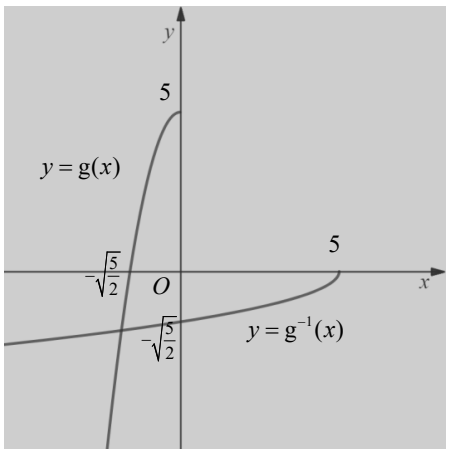
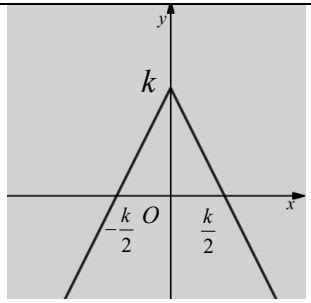
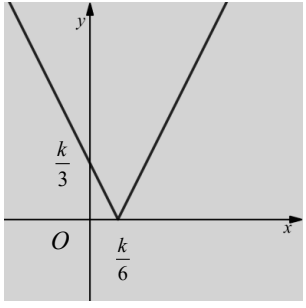
1(a)	$\left(\frac{dy}{dx} =\right) 2x\cos\left(\frac{1}{2}x\right) - \frac{1}{2}x^2\sin\left(\frac{1}{2}x\right)$	M1 A1
	Sets $\frac{dy}{dx} = 0 \Rightarrow \tan\left(\frac{1}{2}x\right) = \frac{4}{x} \Rightarrow x = 2\arctan\left(\frac{4}{x}\right)$*	**d**M1 A1*
		(4)
(b)	$x_2 = 2\arctan\left(\frac{4}{2}\right)$	M1
	$x_2 =$ awrt 2.214 $\qquad$ $x_6 = 2.155$ cao	A1, A1
		(3)
		(7 marks)

2(a)	States or uses $\sin 2x = 2\sin x \cos x$	B1
	$\dfrac{1-\cos 2x}{2\sin 2x} = \dfrac{1-\left(1-2\sin^2 x\right)}{4\sin x\cos x} = \dfrac{2\sin^2 x}{4\sin x\cos x} = \dfrac{1}{2}\tan x$	M1 A1
		(3)
(b)	$\dfrac{9\left(1-\cos 2\theta\right)}{2\sin 2\theta} = 2\sec^2\theta \Rightarrow 9\times"\dfrac{1}{2}"\tan\theta = 2\sec^2\theta$	M1
	Attempts $1+\tan^2\theta = \sec^2\theta \Rightarrow 4\tan^2\theta - 9\tan\theta + 4 = 0$	**d**M1 A1
	$\Rightarrow \tan\theta = \dfrac{9\pm\sqrt{17}}{8}\left(1.64038..., 0.60961...\right)$	M1
	$\Rightarrow \theta =$ awrt $31.4°, 58.6°$	A1 A1
		(6)
(b) Way 2	$\dfrac{9\left(1-\cos 2\theta\right)}{2\sin 2\theta} = 2\sec^2\theta \Rightarrow 9\times"\dfrac{1}{2}"\tan\theta = 2\sec^2\theta$	M1
	$\dfrac{9}{2}\tan\theta = 2\sec^2\theta \Rightarrow \dfrac{9\sin\theta}{\cos\theta} = \dfrac{4}{\cos^2\theta} \Rightarrow 9\sin\theta\cos\theta = 4 \Rightarrow \sin 2\theta = \dfrac{8}{9}$	**d**M1A1
	$\sin 2\theta = \dfrac{8}{9} \Rightarrow 2\theta = 62.7333..., 117.266... \Rightarrow \theta = ...$	M1
	$\Rightarrow \theta =$ awrt $31.4°, 58.6°$	A1 A1
		(9 marks)

3(i)	$\int \frac{12}{(2x-1)^2}\,dx = \frac{-6}{(2x-1)}(+c)$	M1 A1
		(2)
(ii)(a)	(a) $\frac{4x+3}{x+2} = 4 - \frac{5}{x+2}$	M1, A1
		(2)
(ii)(b)	(b) $\int \frac{4x+3}{x+2}\,dx = \int 4 - \frac{5}{x+2}\,dx = 4x - 5\ln\lvert x+2\rvert\ (+c)$	M1 A1ft
	Hence $\int_{-8}^{-5} \frac{4x+3}{x+2}\,dx = (-20 - 5\ln 3) - (-32 - 5\ln 6)$	dM1
	$= 12 + 5\ln 2$	A1
		(4)
		(8 marks)

4(a)	$\text{fg}(x) = \dfrac{4\left(5-2x^2\right)+6}{5-2x^2-5}$		M1
	$\text{fg}(x) = 3 \Rightarrow \dfrac{26-8x^2}{-2x^2} = 3 \Rightarrow x = -\sqrt{13}$		**d**M1 A1 A1
			(4)
(b)	$y = \dfrac{4x+6}{x-5} \Rightarrow yx-5y = 4x+6 \Rightarrow yx-4x = 5y+6$		M1
	$\Rightarrow x = \dfrac{5y+6}{y-4}$		A1
	$\Rightarrow \text{f}^{-1}(x) = \dfrac{5x+6}{x-4} \qquad x \in \mathbb{R},\ x \neq 4$		A1
			(3)
(c)		Shape and position for $y = \text{g}(x)$. A parabola starting on the y-axis that extends down in quadrant 2 to at least the x-axis. Condone slips of the pen e.g. if the curve heads back towards the y-axis in quadrant 3 slightly.	B1
		Shape and position for $y = \text{g}(x)$ as above that crosses the x-axis **and** $y = \text{g}^{-1}(x)$ a parabola starting on the x-axis with graphs crossing as shown in quadrant 3. Condone slips of the pen e.g. if $\text{g}^{-1}(x)$ heads back towards the x-axis in quadrant 3 slightly.	B1
		Correct intercepts. This mark is for the 4 correct intercepts as shown. Allow as values or coordinates and allow coordinates the wrong way round (e.g. (0, 5) for (5, 0)) as long as they are in the correct places. The negative intercepts must be exact as shown or equivalent. The graphs must cross the axes at these points and you can ignore any other intercepts.	B1
			(3)
			(10 marks)

5(a)	States or implies that $\log_{10} p = 0.32$ **or** $\log_{10} q = \dfrac{0.56-0.32}{8}$	M1
	$p =$ awrt 2.089 **or** $q =$ awrt 1.072	A1
	States or implies that $\log_{10} p = 0.32$ **and** $\log_{10} q = \dfrac{0.56-0.32}{8}$	M1
	$p =$ awrt 2.089 **and** $q =$ awrt 1.072	A1
		(4)
(b)	States or implies that $\dfrac{\mathrm{d}A}{\mathrm{d}t} = p \ln q \times q^{t}$ with their values for p and q	M1 A1
	Rate of increase in pond weed after 6 days is 0.22 (m^2/day)	A1
		(3)
		(7 marks)

6(a)(i)		Shape	M1
		Correct intercepts.	A1
(ii)		Shape	M1
		Correct intercepts.	A1
			(4)
(b)	Attempts to solve either $k+2x=-2x+\frac{k}{3}$ **or** $k-2x=2x-\frac{k}{3}$		M1
	Achieves either $x=-\frac{1}{6}k$ **or** $x=\frac{1}{3}k$ oe		A1
	Attempts to solve both $k+2x=-2x+\frac{k}{3}$ **and** $k-2x=2x-\frac{k}{3}$		M1
	Achieves either $x=-\frac{1}{6}k$ **and** $x=\frac{1}{3}k$		A1
			(4)
			(8 marks)

7	$x = 6\sin^2 2y \Rightarrow \dfrac{dx}{dy} = 24\sin 2y\cos 2y$	M1 A1
	Attempts to use $\dfrac{dy}{dx} = 1 \div \dfrac{dx}{dy}$	M1
	Attempts to use both $\sin 2y = \sqrt{\dfrac{x}{6}}$ and $\cos 2y = \sqrt{1-\sin^2 2y} = \sqrt{1-\dfrac{x}{6}}$	M1
	$\dfrac{dy}{dx} = \dfrac{1}{24\sin 2y\cos 2y} = \dfrac{1}{24\times\sqrt{\dfrac{x}{6}}\times\sqrt{1-\dfrac{x}{6}}} = \dfrac{1}{4\sqrt{6x-x^2}}$	A1
		(5)
Way 2	$x = 6\sin^2 2y = 3-3\cos 4y \Rightarrow \dfrac{dx}{dy} = 12\sin 4y$	M1A1
	Attempts to use $\dfrac{dy}{dx} = 1 \div \dfrac{dx}{dy}$	M1
	Attempts to use $\sin 4y = \dfrac{\sqrt{6x-x^2}}{3}$	M1
	$\dfrac{dy}{dx} = \dfrac{1}{12\sin 4y} = \dfrac{1}{12\times\dfrac{\sqrt{6x-x^2}}{3}} = \dfrac{1}{4\sqrt{6x-x^2}}$	A1
Way 3	$x = 6\sin^2 2y = 24\sin^2 y\cos^2 y \Rightarrow \dfrac{dx}{dy} = 48\sin y\cos^3 y - 48\sin^3 y\cos y$	M1A1
	Attempts to use $\dfrac{dy}{dx} = 1 \div \dfrac{dx}{dy}$	M1
	$\dfrac{dy}{dx} = \dfrac{1}{48\sin y\cos^3 y - 48\sin^3 y\cos y} = \dfrac{1}{48\sin y\cos y\left(\cos^2 y - \sin^2 y\right)}$ $\dfrac{dy}{dx} = \dfrac{1}{24\sin 2y\cos 2y} = \dfrac{1}{24\times\sqrt{\dfrac{x}{6}}\times\sqrt{1-\dfrac{x}{6}}}$	M1
	$= \dfrac{1}{4\sqrt{6x-x^2}}$	A1
		(5 marks)

8(a)	200	B1
		(1)
(b)	600	B1
		(1)
(c)	$500 = \dfrac{600\mathrm{e}^{0.3t}}{2+\mathrm{e}^{0.3t}} \Rightarrow 100\mathrm{e}^{0.3t} = 1000, \Rightarrow \mathrm{e}^{0.3t} = 10$	M1, A1
	$\Rightarrow t = \dfrac{\ln 10}{0.3} = 7$ years 8 months	dM1 A1
		(4)
(d)	$N = \dfrac{600\mathrm{e}^{0.3t}}{2+\mathrm{e}^{0.3t}} \Rightarrow \dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{\left(2+\mathrm{e}^{0.3t}\right)\times 180\mathrm{e}^{0.3t} - 180\mathrm{e}^{0.3t}\times\mathrm{e}^{0.3t}}{\left(2+\mathrm{e}^{0.3t}\right)^2}$	M1 A1
	$\Rightarrow \dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{360\mathrm{e}^{0.3t}}{\left(2+\mathrm{e}^{0.3t}\right)^2}$	A1 **(3)**
(e)	$8 = \dfrac{360\mathrm{e}^{0.3t}}{\left(2+\mathrm{e}^{0.3t}\right)^2} \Rightarrow 8\left(\mathrm{e}^{0.3t}\right)^2 - 328\left(\mathrm{e}^{0.3t}\right) + 32 = 0$	M1
	$\mathrm{e}^{0.3t} = \dfrac{41+\sqrt{1665}}{2}$	dM1
	$(t=)\dfrac{\ln\left(\dfrac{41+\sqrt{1665}}{2}\right)}{0.3}$	ddM1
	$(T=)$ awrt 12.4	A1
		(4)
		(13 marks)

9(a)	$R = 13$	B1
	$\tan\alpha = \frac{5}{12} \Rightarrow \alpha =$ awrt 0.395	M1A1
		(3)
(b)	$g(\theta) = 10 + 13\sin\left(2\theta - \frac{\pi}{6} - 0.395\right)$	
(i)	(i) Minimum value is -3	B1 ft
(ii)	$2\theta - \frac{\pi}{6} - 0.395 = \frac{3\pi}{2} \Rightarrow \theta =$ awrt 2.82	M1 A1
		(3)
(c)	$h(\beta) = 10 - 169\sin^2(\beta - 0.395)$	
	$-159 \leqslant h \leqslant 10$	M1 A1
		(2)
		(8 marks)